23 An observer is standing on a railway platform. A train passes the observer at constant speed while emitting sound of constant frequency *f* from its whistle.

What does the observer hear?

A sound of a decreasing frequency as the train approaches and of an increasing frequency as it moves away

B sound of a higher frequency than *f* as the train approaches and of a lower frequency than *f* as the train moves away

C sound of a lower frequency than *f* as the train approaches and of a higher frequency than *f* as the train moves away

D sound of an increasing frequency as the train approaches and as it moves away